

The Urban Overload Hypothesis and Policing: How Call Volume and
Time affect Police Interactions

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Abstract

Police behavior and escalatory responses have become central to political discourse in the United States over the last five years, with many departments under threat of losing funding. The departments that have come under scrutiny have been largely urban, with police having to monitor swaths of area in large cities for 24 hours a day. This begs the question, how are police officers in urban environments affected by the number of calls for assistance they receive, and the time of day that they receive them? This is known as the urban overload hypothesis, which posits that individuals in urban environments are more prone to sensory overload (and thus altered decision making) by the sheer number of stimuli present in their environment. This study ultimately aims to further identify which aspects of police-civilian interactions could contribute to this hypothesis, and how much of an effect each covariate has on the outcome of a situation. Several steps are taken: incidents are categorized, time-to-event data is organized, and regressions are utilized in order to display the probability of arrest for a given incident, taking into account situational factors. The Cox PH model is able to succinctly provide the difference in arrest likelihood for event type and time of day, while multinomial logistic regression displays the contributions of each variable to the possibility of punitive outcomes. Overall, the effects of call volume leave additional avenues for exploration, while evidence of a strong correlation between temporal factors and punitive outcomes remain in the results of the statistical analyses.

1 Introduction

Urban policing operates under a complex set of demands shaped by both spatial and temporal pressures. In high density areas where service calls can fluctuate drastically over short time periods, understanding the decision making processes of officers requires accounting for more than just crime type or severity. Instead, broader environmental factors, such as call volume, time of day, and competing priorities, can shape both the timing and outcome of police interactions (Duxbury et al., 2015). This paper explores these dynamics through the lens of the urban overload hypothesis, focusing the gaze of this broad theory onto police officers.

In the overall context of policing, the urban overload hypothesis translates into officers prioritizing certain calls or individuals based on time constraints and perceived importance (McCauley, 1970). While this theory has often been applied in sociological discussions of urban life, its empirical testing within modern policing contexts has remained limited. This study seeks to contribute to that literature by empirically examining how temporal factors and environmental stressors affect the likelihood and timing of punitive police actions.

To do this, a dataset of police incident-level records San Francisco, California (City and County of San Francisco, 2025), are analyzed using two statistical modeling approaches. The first to be employed is a logistic regression model, estimating the probability that an interaction results in punitive action, and assessing the odds ratios to gain insight into which variables have a significant effect on punitive events. Next, a Cox proportional hazards model is employed to assess how covariates such as call volume, time of day, and incident priority influence the duration from police arrival on scene to the closing of a call involving punitive action (e.g., arrests, citations). These complementary models provide insight not only into whether officers are more or less likely to take punitive actions, but also how quickly such decisions are made in relation to external pressures.

The analysis focuses in particular on temporal stressors: whether police behavior differs during high demand periods, at night, or in the presence of large numbers of concurrent calls. This allows for a direct test of the urban overload hypothesis by examining whether increased environmental demands result in differential treatment of calls, especially those deemed lower priority. By stratifying models across different incident types (e.g., violent crime, public disturbances, mental health calls), we investigate whether the influence of temporal and environmental factors varies depending on the nature of the incident.

While the results of the models are not meant to infer causality due to limitations with the data (factors like geolocation, officer shift data, and the personal details/demographic data of officers and civilians are not known), this information is meant to serve as a baseline to get an insight into how police

behavior plays a role in the probability of arrest. The biggest factors will be how this effect changes based on call volume and temporal factors (difference between police behavior during day and night) and the degree to which the probability of punitive action (citation, arrest, or police report) contribute to specific events.

This research aims to shed light on how structural pressures within urban policing environments shape discretionary decision making. By moving beyond individual or situational explanations, it contributes to a growing body of literature that emphasizes the importance of system level factors in understanding police behavior. The findings have implications not only for academic theory but also for policy discussions about resource allocation, dispatch prioritization, and strategies for mitigating bias and inefficiency in law enforcement responses under pressure.

2 Literature Review

The dynamics of police responses during varying call volumes have significant implications for law enforcement practices, community safety, and relations. As such, this literature review examines the circumstances in which temporal factors and high call volume could potentially affect the behavior of on-duty officers. When Stanley Milgram proposed the urban overload hypothesis in 1970, he suggested that environments experience sensory overload due to constant stimuli, leading them to adopt mechanisms like social withdrawal to manage the overwhelming input ([McCauley, 1970](#)). Applying this concept to policing, this review explores whether officers, when faced with high dispatch activity, are more likely to escalate or prolong incidents compared to low activity periods, and whether specific times of day or incident types correlate with such escalations. The urban overload hypothesis posits that the sensory saturation inherent in urban settings compels residents to limit their social interactions as a coping strategy ([McCauley, 1970](#)). In the context of policing, officers operating in high density urban areas may experience similar overload due to elevated call volumes and diverse situational demands. The thought here is that constant exposure to numerous and varied incidents could lead to cognitive and emotional strain, potentially influencing decision making processes and increasing the propensity for escalatory responses. This idea is addressed in [Duxbury et al. \(2015\)](#), which suggests that repeated exposure to such stimuli could cause something called "work-role overload," potentially causing excess stress, thus inhibiting officers from fully or properly carrying out their duties. Additionally, [Duxbury et al. \(2015\)](#) study posits that police officers who come under more stress (including those with more duties as well as night shift workers) were much more likely to escalate their individual responses, indicated by an uptick in use of deadly force. Despite these discouraging findings, [Mummolo \(2017\)](#) is able to show that even relatively minor changes in officer policy are often able to decrease total interactions between themselves

and citizens, such as more rigorous requirements for officers to use "stop and frisk" tactics. As such, it stands to reason that if cognitive overload is caused by a high number of interactions in an urban environment, it could be counteracted by simply decreasing the ability for police officers to interact with citizens, which is why call volume (reflecting the amount of impending interactions) is utilized throughout the analysis.

Research indicates that increased police workload can affect discretionary practices. The "overload hypothesis" suggests that as workload intensifies, officers may allocate less time and attention to new cases, especially minor offenses (Schulenberg, 2003). This resource strain might prompt officers to resolve incidents more swiftly, possibly through escalated measures such as arrests, to manage their workload efficiently. A study examining police with young offenders found that higher workloads correlated with a greater likelihood of formal actions, like arrests, as opposed to informal resolutions (Schulenberg, 2003). This trend suggests that during peak call periods, officers might resort to more definitive actions to expedite case closure, thereby diminishing the cumulative effect of pending incidents. Another example of this phenomenon is given by Mummolo (2017), who studied officer behavior before and after the implementation of tightened restrictions surrounding "stop and frisk" policies in New York City. Once restrictions were set in place, police interactions decreased, as did escalatory responses, such as the rate of improper detainments, showing a potential link between the amount of interactions and subsequent rates of police escalation (Mummolo, 2017).

Another factor that plays a crucial role in potentially escalatory responses is the timing of an incident. Research has shown that increased night patrols, as opposed to day patrols, can enhance police effectiveness (Rouse, 1985). This suggests that nighttime incidents might receive more proactive attention, potentially leading to different escalation patterns compared to daytime incidents. Approaching this idea from a biological perspective, James et al. (2018) cites that night time workers often suffer from excess fatigue due to their shifts failing to align with the natural circadian rhythm of the human body, potentially causing difficulties with alertness and decision making. This compromise between the efficacy of the patrols themselves versus the mental state of individual officers on nighttime shifts appears to be an unexplored avenue, leaving room to investigate this further. Additionally, the nature of incidents is another factor that potentially influences the likelihood of escalation. High priority calls, such as those involving violent crimes, inherently demand more immediate and assertive responses. During periods of high call volume, however, even low priority incidents might be handled differently than normal, especially if officers perceive a need to quickly resolve situations to address subsequent issues, or are changing shifts (DeAngelo, 2023). These subtle but important temporal shifts are key to identifying changes in major variables, such as event times, and implicating the effects of call volume on workload.

The cognitive load experienced by officers during high call volumes can impair decision making

processes. [Mugford \(2013\)](#) emphasizes the importance of improving police training from a cognitive load perspective, suggesting that better training can help officers manage the stress and complexity of high demand situations. Revisiting [Duxbury et al. \(2015\)](#), the array of roles that police officers are asked to perform on a daily basis can often cause them stress through what are known as "competing demands." This refers to a given officer's tasks, which can range from field work, to paperwork, to interactions within the judicial system ([Duxbury et al., 2015](#)). As such, the multifaceted nature of police officers' jobs often adds complexity to their routines, which according to [Duxbury et al. \(2015\)](#), adds difficulties to focusing on tasks at hand. [Duxbury et al. \(2015\)](#) also finds that the culture in many law enforcement agencies discourages officers from saying "no" to additional assignments, even when they are aware that added stressors could potentially decrease their performance. This comes at a time when performance among police officers is at a premium, as [DeAngelo \(2023\)](#) informs us that many metro area law enforcement agencies across the US are facing funding cuts. This emphasizes the importance of decreasing officers' "competing demands" and increasing their availability, as the results of [DeAngelo \(2023\)](#) show that a 10% increase in response times to violent crimes reduces the likelihood of reported injury by 4.3%. This strongly indicates the need for analysis into volumetric factors that correlate with, or potentially reduce officer demands in the name of public safety.

Approaching police-civilian interactions from an ecological perspective, [Klinger \(1997\)](#) suggests that the environment in which officers operate significantly influences their behavior. In high-activity periods, the ecological pressures of increased incidents and limited resources may lead officers to adopt more assertive tactics as a means of maintaining control and order. This perspective aligns with the urban overload hypothesis, as both suggest that environmental factors critically shape individual responses. Additionally, it may be important to view temporal factors ecologically throughout the course of this project, as the physical environment of the night or day fundamentally changes how police officers are able to respond to situations. [Rouse \(1985\)](#) posits that physical obstructions (combined with darkness) often make it more difficult for officers to spot criminals at night time, but that the lack of other citizens also makes it easier for officers to spot suspicious movement. These factors point to officer behavior potentially changing as a result of their environment, potentially affecting response times or interaction times. The connections between high call volumes and escalatory police responses are multifaceted, involving cognitive, temporal, and environmental factors. The urbanoverload hypothesis provides a solid framework for understanding how sensory and informational overload in urban policing can alter response times, prolong incidents, or even change arrest rates. Addressing these challenges requires holistic analysis into the contributing factors of police overload, reflecting on the extent to which it is a factor, and employing statistical analysis in identifying the highest leverage situational traits.

3 Methodology

3.1 Research Question and Study Design

This study investigates how temporal factors and high call volume influence police interactions in San Francisco, as recorded in dispatch data. Specifically, the study examines how these factors shape the type of call received, the disposition of the call, and time spent on scene. The research is guided by the urban overload hypothesis, which suggests that high environmental stimuli in urban areas lead to more reactive, less deliberative responses by public service agents, including police.

This research is non-experimental and relies on observational data collected from San Francisco’s public dispatch records. Because the data captures real-world police dispatches, the study cannot establish true causality but can identify significant associations between temporal and volume factors and key outcomes. While a true experimental design could better establish causality, such designs are infeasible given the nature of emergency response systems. Therefore, this study employs event history analysis (hazard regression) to model how external factors such as call priority, time of day, and call volume influence the duration of police interactions, coupled with regression modeling for categorical outcomes (call type and disposition).

This non-experimental longitudinal design allows for modeling time-to-event-data (in this case, the duration of police response) from dispatch to completion, while also capturing variations due to shifting temporal and environmental pressures.

The primary dataset is San Francisco Police Department Computer-Aided Dispatch (CAD) data, available through San Francisco’s Open Data Portal ([City and County of San Francisco, 2025](#)). This dataset includes detailed logs of police dispatches, capturing timestamps, call types, dispositions, response durations, call priority (before and after), call sensitivity, and location data.

3.2 Analytical Methods

3.2.1 Event History Analysis (Hazard Regression)

To analyze time on scene, the study will employ a Cox Proportional Hazards Model. This model is appropriate for time-to-event data, or "event history data structures," ([Box-Steffensmeier and Jones, 2004](#)) which, in this case, will signify the completion of the police interaction, as call duration is being predicted. Temporal and call volume variables are incorporated as time-varying covariates, and the expected output of this model would allow it to capture how changing call loads or time of day dynamically shape scene duration of law enforcement interactions.

This type of event history analysis has been conducted in similar articles, such as in [Li et al. \(2015\)](#),

which employed a log-logistic competing risks mixture model, utilizing time-to-event data containing the length of traffic incidents in Singapore. The study notes, however, that there could be potential issues with the log-logistic distribution’s characterization of the data (Li et al., 2015). For this reason, to analyze the police dispatch time-to-event dataset, several Cox Proportional Hazards regressions will be used for each interaction type, as the semi parametric approach of this model type only makes parametric assumptions about the predictors, but does not assume the shape of the function itself (e.g. log-logistic) (Harrell, 2015).

Another major reason for the employment of such a model for looking into time-to-event-data is its ability to handle ”censored,” or missing data. Within event history data structures, it is common for individual events to be split in half or ”censored,” at the beginning of recordkeeping (left censoring) or at the end (right censoring) (Box-Steffensmeier and Jones, 2004), which is a potential issue. However, Cox PH regressions allow for avenues to incorporate likelihood functions for censored data, allowing these observations to add information on ”survival” or the function outcome, but not the time elapsed (Box-Steffensmeier and Jones, 2004).

Further factors contributing to the choice of a Cox PH regression is its ability to incorporate time varying predictors such as call volume spikes into analysis. Cox Proportional Hazards regressions are often used in disease progression analysis, as Fisher and Lin (1999) gives advocates for the use of these covariates in situations where they do not create a self fulfilling prophecy (such as a person dying from a disease). In the case of this study, such covariates would be applicable, as the outcome of individual events or interactions are not certain.

3.2.2 Multinomial Logistic Regression for Call Disposition

For modeling disposition (the outcome of the call), a multinomial logistic regression will assess how time and volume factors shift the likelihood of punitive outcomes (citation, arrest, reprimands). Revisiting Li et al. (2015), the study employs logistic regression in its prediction of a categorical variable that illustrates the outcome of dispatched traffic incidents. In turn, utilizing odds ratios to interpret how individual factors contribute to the probability of punitive action is key to gaining insight into basic assumptions about the data and building additional models. Further, using this method with police dispatch outcomes can give insight into individual predictors, revealing the likelihood of an event occurring, or by also potentially illustrating the relative risk, or the ratio of two possible outcomes (Hosmer et al., 2013). The importance of this method ultimately lies in its ability to address individual contributing factors in the course of a punitive event, contributing the analysis process largely through properties of exploration and Cox PH model building.

3.3 Assumption Checking and Validation

3.3.1 Cox Proportional Hazards

A key factor in checking model assumptions for a Cox PH regression are utilizing the Schoenfeld residuals to check goodness of fit. [Xue and Schifano \(2017\)](#) describe the Schoenfeld residuals as being checked against expected values at the end of every event, given the risk at that point in time, which is essential for checking what is known as the proportional hazards assumption. The proportional hazards assumption means that if predictor variables do not change over time, then the ratio comparing two observations is constant ([Xue and Schifano, 2017](#)). Ultimately, a graph of the Schoenfeld residuals will be scattered around zero if the proportional hazards assumption is met ([Xue and Schifano, 2017](#)).

Literature by [Klein and Moeschberger \(2003\)](#), talks about the importance of the score test for association of population subgroups, and how it is utilized. This test is necessary in the context of this study, as potential associations between individual event outcomes (interaction time, etc.) could ultimately be tied to unobservable demographic variables not present in the dataset. Searching for potential associations between the times of events occurring is key, as this study plans to employ multiple Cox PH regressions for each event type, therefore a p-value below 0.05 on the score test would indicate that such an association exists ([Klein and Moeschberger, 2003](#)).

Additionally, AIC can also be used with Cox Proportional Hazards models. [Harrell \(2015\)](#) indicates that AIC indicates partial probability in the context of Cox PH models, as an assumption of these models is that the baseline hazard (when covariates are equal to zero) is unspecified. With this in mind, [Harrell \(2015\)](#) use of AIC suggests that it is a predictor of covariate efficacy in this context rather than its typical role of describing model fit.

3.3.2 Multinomial Logistic Regression

To check for violations of assumptions, eliminating variables that are potentially multicollinear, examining outcome coefficients against for significance and, and checking the independence of errors graphically by plotting residuals over time ([Stoltzfus, 2011](#)). For robustness, a sensitivity analysis will be undertaken by subsetting the data into 70% (testing) and 30% (training) sections, also making effort to exclude outliers and high leverage cases ([Stoltzfus, 2011](#)). Additionally, ten fold cross-validation will be conducted to estimate out of sample performance ([Stoltzfus, 2011](#)). Finally, to validate the multinomial logistic regression model, fit will be assessed using the likelihood ratio test ([Hosmer et al., 2013](#)).

Table 1: Conceptual Model and Research Details

Category	Details
Research question	How do temporal factors (time of day, day of week, seasonal trends) and high call volume influence police interactions in San Francisco, as recorded in dispatch data?
Study Design	Non-experimental, observational study using secondary data from San Francisco Police Department’s Computer-Aided Dispatch (CAD) system. Longitudinal analysis focusing on time-to-event data and categorical outcomes.
Primary Data Source	San Francisco Police Department Computer-Aided Dispatch data. Includes timestamps, call types, dispositions, response durations, call priority, sensitivity, and location data.
Analytical methods	Event History Analysis (Cox Proportional Hazards Regression): Models time on scene from dispatch to completion. Incorporates temporal factors (time of day, day of week) and call volume as time-varying covariates. Multinomial logistic regression: Models the disposition (outcome) of calls, predicting likelihood of different outcomes based on temporal and volume-related factors.
Assumption checks and diagnostics	Cox PH Model: Check proportional hazards assumption using Schoenfeld Residuals; perform score test for population subgroup associations; evaluate predictive efficacy with AIC. Multinomial logistic regression: Check for linearity between predictors and log-odds, and use likelihood ratio test for overall fit. Ten fold cross validation for robustness.
Expected outcomes	Identify time periods associated with longer or shorter police interactions. Determine whether higher call volumes are linked to faster scene clearance or longer scene durations. Assess how temporal and volume factors shift the likelihood of different call dispositions (report taken, resolved on scene, no action, etc.).
Conceptual framework	Guided by the urban overload hypothesis, which posits that urban environments with high external stimuli (such as heavy call volume) lead to more reactivity by police.
Causal limits	Observational data limits causal inference, but potential strong associations between time, volume, and key outcomes can be informative.

4 Data

4.1 Dataset and Use

The usage policy for the data comes from the Open Data Commons’ Public Domain and Dedication License. This means that the dataset is able to be shared, modified and reused without restrictions ([City and County of San Francisco, 2025](#)).

4.2 Data Cleaning

The first step taken in cleaning the data was removing passing calls that do not represent actual emergency responses. Next, calls with missing location information due to their sensitive nature were removed

as well (e.g., juvenile cases, sexual cases of all ages). Time variables were created for night, incident length, and call volume to be used later in analysis. NA rows and variables not pertinent to the analysis were also taken out of the data, with examples such as "data_loaded_time". The overall analysis included data from 2022 onwards. This subset of time was used due to changes in recordkeeping within the dataset itself between pre and post 2022, as well as to account for new potential biases present in a post-covid, post Minneapolis riot era.

4.3 Data Set Components

The data set containing the dispatch data is mainly temporal, but also includes spatial components. While the call times for dispatch, and the resolution and contact times for the officers are the main focus, for non sensitive cases there is often a location component included in each entry. This could be an intersection, coordinates, or even hinted at by the police district. Additionally, no personal information regarding those involved in specific incidents is present, including dispatchers, police officers, or civilians. New variables were created include night flag (indicating 1 for night and 0 for day), as well as call volume variables. Night flag is set to "1" for all incidents where 80% or more of the elapsed time occurred between 21:00 and 4:00 (9pm and 4am), while call volume reflects the number of concurrent calls (those that were opened beforehand and finished during or after a given call). The chosen time period for night flag reflects San Francisco's latest sunset and earliest sunrise times of the year, taking into account literature from [Rouse \(1985\)](#), which posits that physical limitations to sight during night hours significantly alters police work. Beyond these variables, all others are used in the analysis (aside from the categorized incident types) are present in the default structure for the data set.

4.4 Ethical Considerations

When working with public service data such as police dispatch data, it is crucial to handle the information ethically to respect both the departments, the communities they serve, and respect both sides of this contentious social topic. Protecting the privacy of individuals involved in the dispatches is essential, so any personally identifiable information must be anonymized or aggregated (though much of it already is, especially in cases involving juveniles as well as sexual abuse cases). It is also crucial be aware of potential biases in the data, such as those related to race, location, or time of day, and ensure these factors do not lead to unjust conclusions. The analysis should be approached with sensitivity to avoid perpetuating stereotypes or reinforcing harmful narratives about police interactions. Additionally, the findings should be presented transparently and responsibly, with clear communication of any limitations to avoid misinterpretation. By prioritizing privacy, fairness, and transparency, the research can contribute

to informed decision making without compromising public trust.

5 Results

The investigatory part of this research looks into how temporal factors and urban overload influence the length of police interactions and their outcomes. Prior to analysis, it was hypothesized that call volume and event type would have significant positive effects on the outcome of an interaction (arrests, citations, police report filed). Methods employed in this analysis include Multinomial logistic regression to model event outcomes (punitive action, no punitive action), and view contributing factors to these events. Additionally, Cox Proportional Hazards regressions are employed for each event category (e.g. violent crime, property crime), assessing how temporal factors and high call volume change rates of punitive action over the duration of an incident.

For this portion of the analysis, only records from 2022 onward were used, as there was an overall decrease in reports included in the data starting at this point, possibly pointing to post covid strategy changes, or policing strategy changes put into effect after nationwide riots against the police. Additionally, only events where punitive action occurred were factored into Cox Proportional Hazards analysis, as time variations for events that did not end in a police report or arrest did not mirror those of events that did, adding unexpected noise to the analysis. As the focus of this analysis is on how police interact with members of the public based on mediating factors, the priority of this undertaking was to ensure that links included in statistical analysis of time-to-event data were relevant police-citizen interactions.

5.1 Multinomial Logistic Regression

In order to measure the contributing factors in event outcomes, a multinomial logistic regression was employed in analysis. The contributions of each variable, reported by the log odds ratio, allows for assessment of how these factors perform in the prediction of whether or not an event results in punitive action. The variables in question represent the minutes an officer spends on scene, the category of the alleged crime reported by dispatch, and the priority of the incident. The priority of the incident is denoted by letters or numbers, letters denoting severity of crimes ("A"- "C", highest to lowest, while "I" refers to minor incidents), numbers indicating medical emergencies ("3"- "1" highest to lowest priority, "E" indicates most severe emergencies suggesting imminent death). Overall, this model aims to confirm baseline assumptions about the hypothesis, such as the connections between call priorities, and call types, while also contributing to variable selection in Cox PH models.

In Table 3, the log odds ratios for each contributing factor are listed. While the minutes on scene and call volume have little effect on the chances of a punitive outcome, the type of crime being perpetrated,

Table 2: Multinomial Regression Accuracy

Accuracy	Kappa
0.7701	0.3674

as well as the resulting priority given to a dispatch call, have a large effect on the overall probability of arrest, police report, or issued citation. The largest of such effects stems from property crimes, which increases the odds of punitive action by police by over five times when compared to other groups.

While the results of the multinomial logistic regression indicate strong ties between event type and chance of punitive action, there is no evidence from this model itself that suggests call volume or interaction length as significant predictors of event outcome. Additionally, call priority and call type are closely intertwined (as each type of call has a specific priority), potentially making the inclusion of both a self fulfilling prophecy. To mitigate this issue, the initial priority of the call is used in the model rather than the final priority, as final priority is nearly always indicative of the call’s outcome. Meanwhile, the kappa values (found in Table 2), as well as the accuracy of the model suggest that it performs significantly better than a null model, but that there are likely limitations from what can be inferred, as a value of 0.367 indicates that predictive power is still somewhat low.

Table 3: Multinomial regression odds ratios

(Intercept)	0.3534209***
mins onscene to close	1.0038469***
call volume	1.0055610***
incident cat Disturbances & Public Order	0.5909586***
incident cat Mental Health & Welfare Checks	0.8012614***
incident cat Property Crimes	5.6354853**
incident cat Violent Crimes	1.7748830***
priority original2	0.6760888***
priority original3	0.6542890***
priority originalA	0.6153811***
priority originalB	0.2959570***
priority originalC	0.7144422***
priority originalE	0.5818688**
priority originalI	0.5695538***
night flag	0.9483487***
** $p < 0.01$, *** $p < 0.001$	

To measure the combined predictive accuracy of the model’s factors, assessing whether or not the model predicts punitive outcomes at a rate greater than chance, refer to Table 2, which displays the accuracy of the model against its predicted values, as well as the kappa statistic. The predicted values from the model are 77% accurate, with a Kappa value that is greater than zero, indicating that the model is, in fact, accurate beyond chance. Despite this, it appears that the call volume does not have a significant effect on the outcome, as the odds ratio for the variable is 1.0056. This indicates that the

model is not finding a connection between call volume and incident outcome in its predictions, despite the variable’s significant p-value. Additionally, the effects of night on call outcome are relatively low as well, slightly decreasing the risk of a punitive outcome. The variables that indicate the highest risk of an increase in citation or arrest are the category of crime being perpetrated, as well as the dispatched priority of the call, exposing a logical link between the priority of a given call and how it is ultimately handled.

5.2 Cox Proportional Hazards Regressions

Cox PH models were employed for further analysis, taking into account call volume as well as temporal factors. Temporal periods (e.g. evening to dawn) are defined in order to capture potential effects of temporal factors on the disposition of the event, as well as modeling time-to-event data. The Cox PH Models are categorized by event type within Kaplan-Meier plots, indicating the differences between each model type and the likelihood of arrest based on the given situation. Additionally, the Schoenfeld residuals were examined via plots (see appendix, Figures 2 and 3), and the proportional hazards assumption was tested utilizing these residuals along with a score test, displayed in Table 4 (Box-Steffensmeier and Jones, 2004).

Table 4: Score Test for Proportional Hazards by Incident Group

Group	Variable	p
Mental Health & Welfare Checks	night flag model	0.13 0.13
Disturbances & Public Order	night flag model	0.14 0.14
Violent Crimes	night flag model	5.9e-08*** 5.9e-08***
Property Crimes	night flag model	1.4e-05*** 1.4e-05***
Other/Uncategorized	night flag model	0.00013*** 0.00013***
*** $p < 0.001$ indicates a violation of PH assumption		

Initially, the model incorporated both the night flag variable as well as the call volume variable to predict the likelihood of arrest. This was done by further cutting down the dataset to only situations that ended in punitive action, as the Cox PH model would otherwise interpret a lack of arrest or citation as right censoring, implying a continuous underlying probability of punitive action, even for those who did not receive punishment (Box-Steffensmeier and Jones, 2004). Additionally, focus on punitive incidents allows insight into interactions that are known to be an officer-citizen interaction, rather than allowing for other noise and uncertainty. The low predictive power of call volume in the multinomial logistic regression holds true for variations of the Cox PH model as well, as initial results incorporating both

the call volume and night flag variables were found to violate model assumptions (see appendix, Table 6). The significant p-value in this case indicates a violation of the proportional hazards assumption, indicating non-linearity of the Schoenfeld residuals (Box-Steffensmeier and Jones, 2004), and suggesting call volume as the culprit, as the variable increases with time rather than remaining constant.

Additional Cox PH regressions were run incorporating only the night flag variable in order to test the temporal components of the hypothesis. The lack of significance for results of the score test in Table 4 suggests that the likelihood of punitive action is reasonably affected by the night flag in situations that include welfare checks and public order situations (as significant values in the context of a Cox PH score test indicate a violation of the model’s assumptions). This finding suggests that the police do interact differently with those who receive a punitive response based on the time of day solely for situations that involve mental health, welfare, disturbances and public order. Meanwhile, miscellaneous calls, along with violent and property crimes, do not meet the assumptions for the Cox PH score test, meaning that these events potentially do carry a constant effect over the life of the event (Klein and Moeschberger, 2003).

Table 5: Cox Model Summary

Group	Measure	Value
Mental Health & Welfare Checks	Coefficient	0.2315
	Hazard Ratio	1.2605
	z-value	9.722
	p-value	2.42e-22***
	AIC	154339.5
Disturbances & Public Order	Coefficient	0.2405
	Hazard Ratio	1.2719
	z-value	12.8024
	p-value	1.59e-37***
	AIC	310305.2
*** $p < 0.001$ indicates significance		

Looking to Table 5, we can see that the p-values of models that passed the score test (disturbances and welfare checks) are both significant. Further, viewing the model from the lens of Box-Steffensmeier and Jones (2004), we can interpret the effect of the coefficients hazard ratios for each event. The coefficients represent the log hazard rate for a one unit increase in the predictor variable, in this case illustrating the difference between night and day. The transformation of the log hazard coefficient is represented by the hazard ratio, which gives insight into relative risk from one group to another (Box-Steffensmeier and Jones, 2004). In this case, the hazard ratio of 1.26 for mental health and welfare, along with a ratio of 1.27 for public order events signifies a 26% and 27% increase in arrest probability, respectively, between day and night for each event type. This is also reflected visually, with the Kaplan-Meier curve displayed in Figure 1 illustrating the increased hazard risk at night for both event types. This not only points to the fact that events tend to end in arrest much more quickly and frequently at night than during the

day.

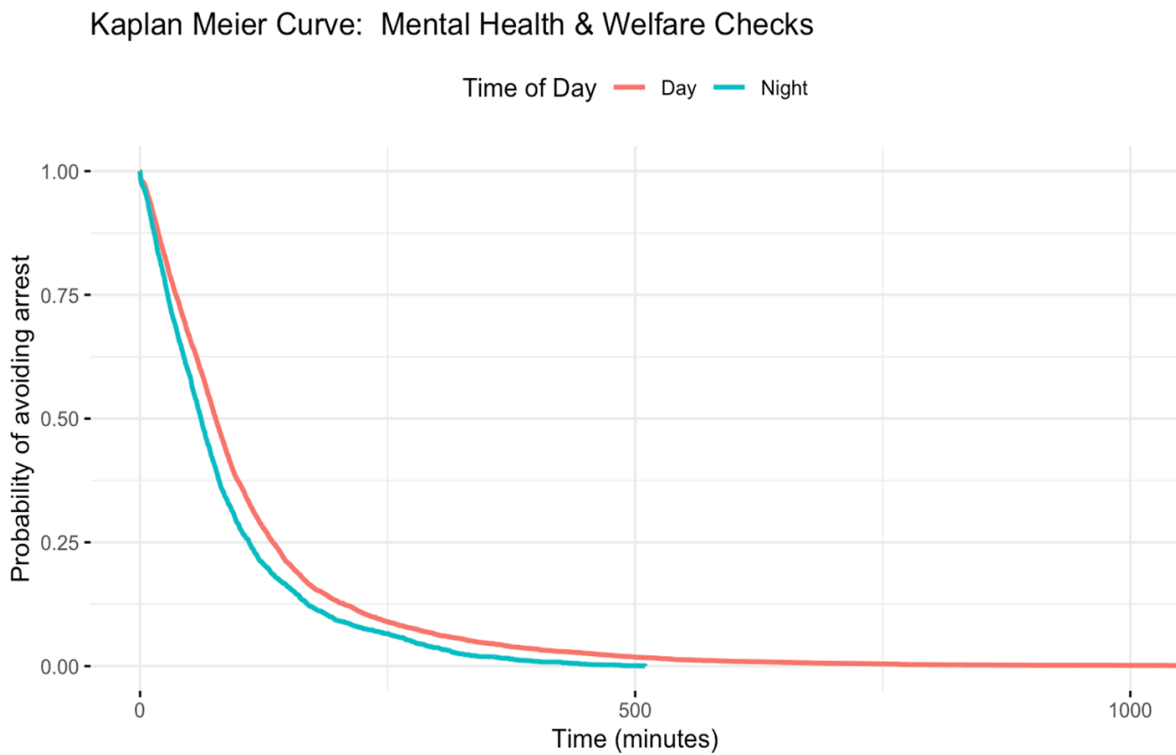


Figure 1: Time to punitive action for mental health and welfare checks, disturbances and public order based on time of day

Overall, the predictive power of the Cox PH regressions for both disturbances and public order, and

mental health and welfare checks appear high. Comparing summary statistics from table 5, the AIC score of the welfare check model is much lower than that of the disturbances, each has a high z-value, which further indicates that the night variable has a significant effect on the hazard probability in these scenarios. Ultimately, the AIC is likely increased for welfare checks due to the specificity of the event type, while the disturbances category contains a larger sample of unique events.

6 Discussion and Conclusion

While the ultimate efficacy of analysis is clear, the main issues resided with incorporating call volume as a variable for analysis. The concept of call priority along with police dispatch's effective use of location services likely played a significant role in this statistical gap. Literature by [Dunnett et al. \(2019\)](#) indicates that modern police dispatch uses both an effective priority system in conjunction with up-to-date location information on all of its officers. This gives potential insight into why call volume has little effect on incident outcome and length, because rather than officers rushing through a lower stakes incident, potentially making a mistake and changing its outcome, officers are sent to priority incidents first, then redirected to more menial tasks ([Dunnett et al., 2019](#)). While it is important to note this lack of immediate response to lesser incidents could also point to disorganization within San Francisco Police Department, the lack of location data for specific officer responses largely inhibits exploration of this avenue. Additionally, SFPD has actually increased its funding for officer salaries and organization tools by over \$200 million over the past five years ([Johnson, 2024](#)), pointing to department policy as a driver for effective dispatch prioritization as opposed to widespread department disorganization.

As exploration into call volume indicated, incident type and resulting priority proved to be key in the overall analysis of punitive action outcomes. The logistic regression reflects that if the call is given specific priority, it is much more likely to result in punitive action, pointing specifically to the odds ratios in Table 3, violent crimes and property crimes are especially likely to result in punitive action, and as such, are given high priority by dispatch. While the kappa value above zero, and the accuracy level of nearly 77% the regression metrics in Table 2, signify the model sees the data at a rate significantly better than chance. However, these findings still leave room for potential further explanation, as demographic variables, officer shifts, and dispatched location data were not present in the data. These factors could add to the model's explanations, potentially giving more specific insight into how incidents were resolved in the context of call volume and temporal factors. As such, the multinomial logistic regression was able to indicate the significance of specific variables, more specifically incident categories, but ultimately fails to provide meaningful insight on the specifics of when punitive action is taken.

While violent crimes and property crimes were not found to meet the score test assumptions for

the Cox PH model, temporal results from mental health checks and public order events proved to vary significantly based on temporal factors. Referring to the odds ratios in Table 5, mental health and welfare, along with disturbances in public order, garnered 26% and 27% increases in arrest likelihood when they occurred at night, which does point to a true correlation between the two factors, though discussion of potential confounds also need to be addressed. For example, while this could point to protocols put in place for police forces, as mentioned in [Dunnett et al. \(2019\)](#), the significance of the models also ties temporal factors to ideas surrounding officer fatigue and the effects of working shift at night. While this is extremely common for most police officers, [James et al. \(2018\)](#) posits that nighttime workers often still suffer from excess fatigue, a result of your body fighting back against your natural circadian rhythm. While this relationship is not considered causal, it could provide insight for SFPD in terms of handling these situations going forward, and illustrate the need to monitor individual differences in the cases to help identify causal factors in the future.

These results underscore the importance of situational context and structural policy when examining policing outcomes under the lens of urban overload and temporal stress. However, the absence of a significant relationship between call volume and punitive action in most models suggests that, despite temporal clustering of events, police responses remain relatively stable. This likely stems from the effective implementation of a triage system within dispatch, aligning with the findings of [Dunnett et al. \(2019\)](#), where priority and spatial data override volume in dictating officer behavior. The slight decrease in punitive action at night despite lack of strong correlation with increased or decreased call lengths suggests differences in either the nature of nighttime incidents, officer decision making under lower visibility and staffing, or possibly broader behavioral or geopolitical dynamics of those involved ([DeAngelo, 2023](#)). The significant result in both regression types signals that this factor warrants further exploration, potentially through interaction effects or more detailed event data. Future research should further explore these dynamics to inform practices that promote effective and equitable law enforcement responses, but overall, enhanced training programs that focus on decision making under pressure could mitigate the tendency toward escalatory responses during peak periods of call activity.

Despite the promise in these results, the correlational nature of this data makes it difficult to directly infer fatigue in nighttime shift workers. As such, follow up studies could be conducted with shift data, comprehensive dispatch location data, and specific officer accounts of events. Such additions could make it possible to infer a more directly causal outcome on the effects of temporal factors on policing, and could allow avenues for working continuously monitored call volume variable, in order to explore this concept using other statistical methods.

Ultimately, while call volume initially appeared promising as a metric for understanding urban overload in policing outcomes, its effects were obfuscated by incident prioritization and dispatch protocols.

The significant correlation between events occurring at night and punitive outcomes suggests a temporal factor at play, but one that may be shaped more by SFPD's institutional practices than by volume stress as a result of volume. These findings highlight the importance of evaluating policing not just by volume or duration, but within the procedural and contextual rules that define modern law enforcement response. Further research incorporating officer shift data, deployment logs, and geolocation data could add to the illustration of how systemic and temporal factors jointly shape outcomes in complex urban policing environments.

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A GitHub

For more information on the project, including code, data, and documentation, visit the [Police Overload GitHub repository](#).

B Supporting visuals

Schoenfeld residuals for score-tested Cox PH models:

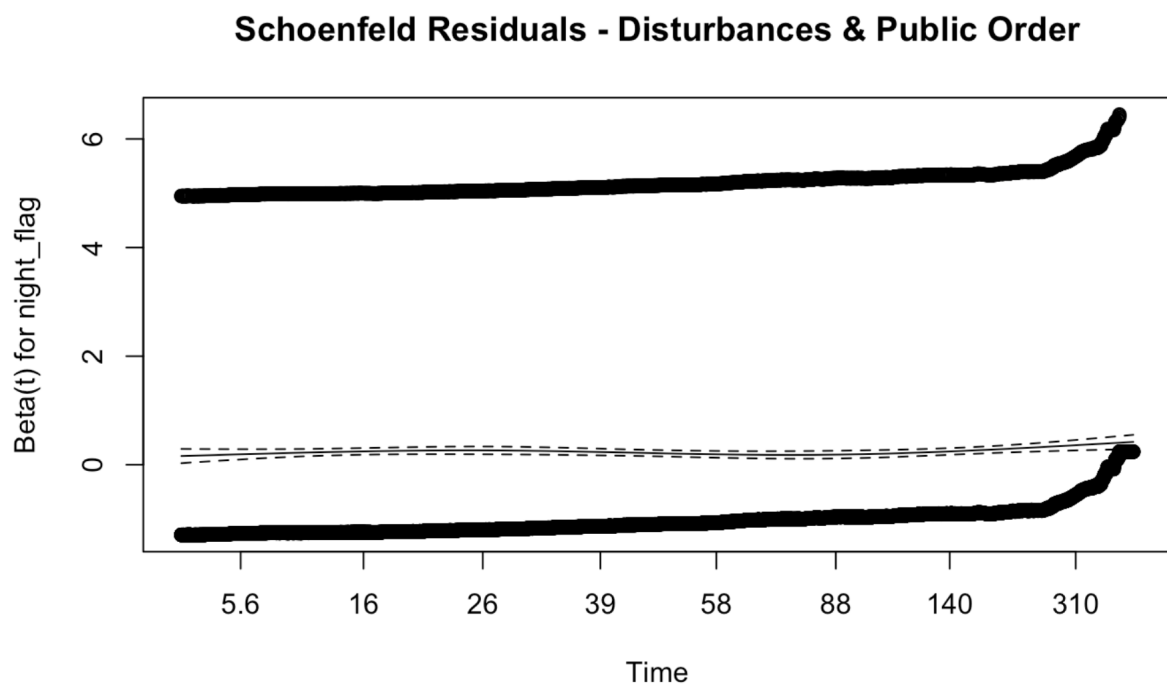


Figure 2: Schoenfeld residuals for disturbances & public order Cox PH model

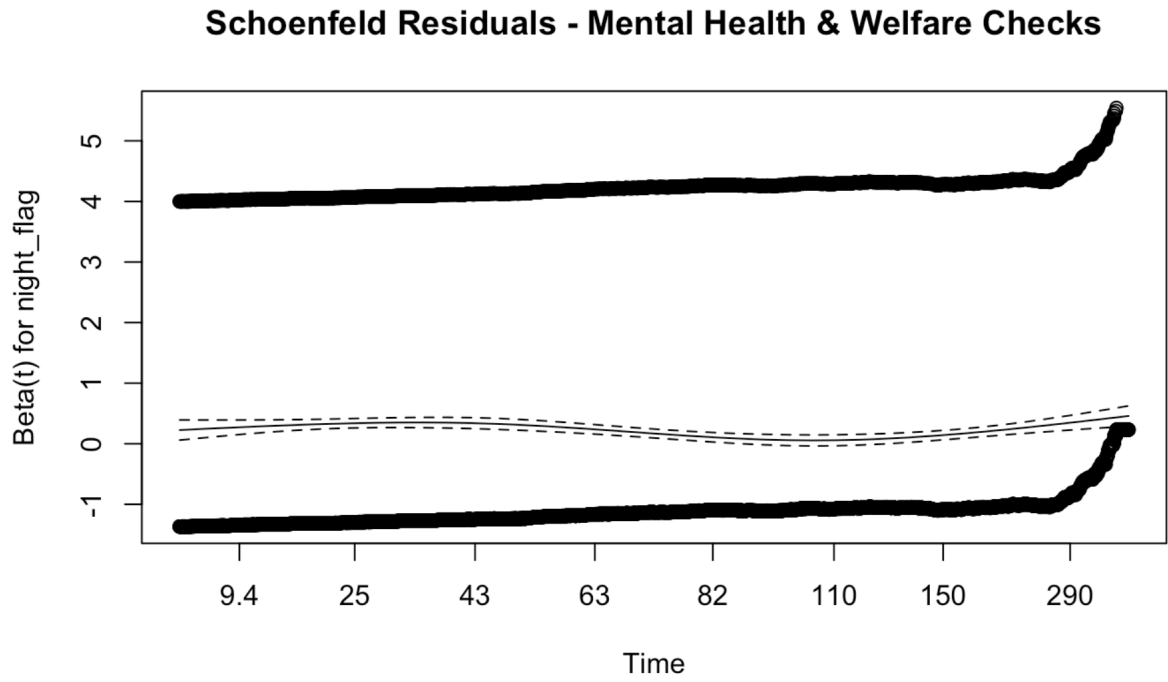


Figure 3: Schoenfeld residuals for mental health & welfare Cox PH model

Table 6: Score Test for Proportional Hazards by Incident Group (Including call volume)

Group	Variable	p
Mental Health & Welfare Checks	night flag	4.3e-08***
	call volume	2e-16***
	model	2e-16***
Violent Crimes	night flag	0.79
	call volume	2e-16***
	model	2e-16***
Other/Uncategorized	night flag	6.7e-08***
	call volume	2e-16***
	model	2e-16***
Property Crimes	night flag	1.4e-12***
	call volume	2e-16***
	model	2e-16***
Disturbances & Public Order	night flag	2e-16***
	call volume	2e-16***
	model	2e-16***

*** $p < 0.001$ indicates violation of PH assumption